Our approach is based on the works of A. Zalizniak. He used "The Roots…" by W.D. Whitney as a data source. The purpose of the approach is to develop a simple and efficient model for Sanskrit morphology. This algorithm is at its core:

I. Choose the root and the form to be generated.

II. Represent a word as a chain of morphemes.

III. Group these morphemes pairwise as “A+B”.

IV. In each pair from the root to the ending:

a. Make alternation regarding Zalizniak's concepts: morphological position from B (the expected grade in A), alternation type of A (the grades A has), and alternation series of A (the set of corresponding basic, guṇa, and vṛddhi grades).

b. Insert a connecting vowel i/ī between A and B if needed (a more refined classification than the traditional seṭ/veṭ/aniṭ was made).

c. Change the alternation series if requested by B;

d. If A is a verbal root or the causative suffix, reduplicate the verbal root if requested by B (six reduplication algorithms were developed).

e. If A is a verbal root and it did not undergo step c, apply verbal root sandhi (phoneme-surroundings dependent changes that occur only in the alternating elements of the verbal roots - as described by Zalizniak).

f. Apply general sandhi (phoneme-surroundings dependent changes that may occur in every type of morpheme - we use M.B Emeneau's algorithm including several additions and exclusions).

The algorithm provides a framework for formulating morphological rules more effectively than other European approaches could offer (e.g., Whitney's rule of sya-future produces 107 exceptions among 777 verbal roots (empirically identified), whereas our rule does not produce any exceptions). Some irregular words (e.g., śvan, yuvan) also became regular when this model is applied.

Currently, we are developing a computer Sanskrit word generator based on this model.